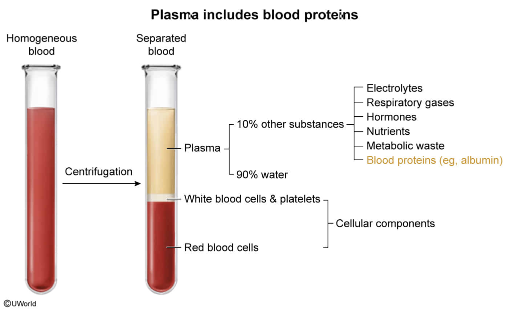
Centrifugation of blood with an anticoagulant agent separates the blood components into layers by density. The least dense layer is the plasma. The plasma layer is most likely to include which component?

- A. Platelets [30%]
- B. Albumin [52%]
- C. Hemoglobin [6%]
- D. Neutrophils [10%]

Correct
52% answered correctly

Explanation:



Blood is the only fluid connective tissue in the body and is composed of living blood cells and nonliving plasma. Centrifugation with an anticoagulant agent separates these components by density, with plasma as the least dense layer. Plasma is the liquid portion of blood and is ~90% water, with the remaining 10% composed of other substances (ie, water, electrolytes, gases, hormones, nutrients, metabolic waste).

One substance found in plasma is blood proteins, such as **albumin**. Because albumin cannot easily cross capillary pores due to its size, it is an important solute for maintaining capillary oncotic pressure (ie, pressure resulting from solute concentration differences between the capillaries and interstitial fluid). The oncotic pressure maintained by solutes such as albumin causes a pulling force for water entry into capillaries via osmosis. This pressure is balanced by the pushing force of blood against capillary walls (hydrostatic pressure), which helps keep fluid in the vasculature.

The substances in the other answer choices would be found in the cellular components of the blood, which originate in the bone marrow:

- Platelets are cell fragments important in the clotting of blood **(Choice A)**.
- Hemoglobin is a protein that binds respiratory gases found within erythrocytes (ie, red blood cells) **(Choice C)**.
- Neutrophils are a type of leukocyte, or cells involved in the immune defense **(Choice D)**.

Educational objective:
Plasma, the liquid portion of blood, is composed of water, electrolytes, gases, hormones, nutrients, metabolic waste, and proteins (eg, albumin).

Time spent:0 sec
Subject:
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